Numerov Method for Bound States in the One-Dimensional Schrödinger Equation

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Contents

1 Introduction

This project studies the one-dimensional time-independent Schrödinger equation using the Numerov method combined with parity-aware shooting procedures. The goals are to validate the method on systems with known analytic solutions, demonstrate numerical convergence and accuracy, and then apply it to more interesting bound-state potentials such as a double well and a finite square well. Extensions to scattering problems are also explored as a secondary component.

This problem is a boundary-value eigenvalue problem rather than a standard initial-value problem: only special energies produce wavefunctions satisfying the required boundary behavior. The Numerov method is well suited to this task because the time-independent Schrödinger equation has the second-order form $\psi'' = q(x)\psi$, with no first-derivative term. The project therefore focuses not only on obtaining eigenvalues, but also on checking that the numerical results are physically meaningful through analytic benchmarks, convergence tests, parity and node structure, and probability conservation in the scattering extension.

The full code was written in Python and is available in this repository. The implementation is modular, with separate components for the Numerov integrator, shooting method, root finding, analysis utilities, and an automated validation suite. This structure allows for independent validation of each numerical component and makes the overall workflow easier to understand.

2 Computational approach

2.1 The Schrödinger equation and Numerov method

In dimensionless units, the Schrödinger equation is written as:

$$\psi''(x) = 2\,[V(x) - E]\psi(x) \tag{1}$$

The wavefunction is integrated with the Numerov scheme on a uniform grid. For symmetric potentials, even and odd states are treated separately using parity conditions at $x = 0$. Eigenvalues are found by scanning for sign changes in a scalar mismatch function and then refining the resulting brackets with robust root finding; in the outward Numerov solver this is bisection followed by a few safeguarded secant-style polishing steps, while the inward-decay and RK4 solvers use straight bisection. The correctness of the numerical approach is verified through comparison with analytical solutions where available, through systematic convergence studies, and through direct automated tests of the implementation.

The overall numerical workflow is implemented in a modular way in the accompanying Python code. The core components include a Numerov integrator for propagating the wavefunction, a shooting module that enforces parity-based boundary conditions and constructs the mismatch function, and analysis utilities for convergence studies and parameter sweeps. This separation of concerns allows each numerical component to be validated independently and makes the implementation easier to interpret [1].

2.2 Shooting and root finding

The boundary-value eigenproblem is reformulated as a root-finding problem for a mismatch function $M(E)$. For a given trial energy E, the Schrödinger equation is integrated and a boundary mismatch is evaluated. The code uses two related formulations. For the infinite square well, finite square well, and quartic double well, the half-domain problem is shot outward from $x = 0$ with parity-consistent startup values and the mismatch is the outer boundary value $M(E) = \psi_E(x_{\max})$. For the harmonic oscillator and the RK4 comparison, the code instead shoots inward from $x_{\max}$ through the decaying tail and enforces parity at the origin:

$$\text{even states:} \quad M(E) = \psi'_E(0), \qquad \text{odd states:} \quad M(E) = \psi_E(0) \tag{2}$$

Eigenvalues correspond to the roots of $M(E) = 0$, which are located by bracketing sign changes and then refined numerically. To make the procedure transparent, we plot $M(E)$ versus E and mark the bisection iterates; zero crossings identify the allowed energies.

In the implementation, the inward even-state mismatch uses a fourth-order backward finite-difference stencil to approximate $\psi'(0)$ from the last few points of the inward integration grid. This is important because a lower-order boundary derivative would spoil the fourth-order behavior otherwise expected from the Numerov recurrence in the harmonic-oscillator study.

3 Results

Each potential is used to validate, test, or demonstrate different aspects of the numerical method, ranging from analytic benchmarks to nontrivial quantum behavior.

3.1 Infinite square well

The infinite square well serves as a benchmark problem with known analytic spectrum:

$$E_n = \frac{(n+1)^2\pi^2}{8a^2} \tag{3}$$

The potential is:

$$V(x) = \begin{cases} 0 & , |x| \le a \\ \infty & , |x| > a \end{cases} \qquad \text{(implemented numerically with a large finite } V_0\text{)} \tag{4}$$

The numerical eigenvalues agree with the analytic values to essentially machine precision, as shown in Fig. 1. In the saved run the first four relative errors are 4.9×10^{-11}, 1.3×10^{-10}, 2.6×10^{-11}, and 7.0×10^{-12}. This is exactly the behavior expected for a benchmark problem of this type: the energies follow the analytic law $E_n \propto (n+1)^2$, and the numerical points are visually indistinguishable from the exact curve on the scale of the plot.

The probability densities for the first four states (Fig. 2) also have the expected structure. The ground-state density is maximal at the center, while the excited states show the correct sequence of interior nodes inherited from the alternating even/odd parity eigenfunctions. All densities vanish at $x = \pm a$, consistent with the hard-wall boundary condition.

The shooting method is validated by plotting the boundary mismatch functions for both even and odd sectors (Fig. 3). The even-sector scan has zero crossings near $E \approx 1.234$ and $E \approx 11.103$, while the odd-sector scan crosses zero near $E \approx 4.935$ and $E \approx 19.739$. These are precisely the first four physical eigenvalues, so the parity separation and bracketing logic are behaving as intended rather than picking spurious roots.

The convergence of the energy error as a function of grid spacing h (Fig. 4) is also as expected for Numerov. The fitted exponents from the saved convergence study are $p = 3.94$, 4.00, 3.98, and 4.00 for the first four states, which is strong evidence that the recurrence, startup values, and boundary treatment are all consistent with fourth-order accuracy. The higher states sit at larger absolute error for a fixed grid, which is also expected because their shorter wavelengths are harder to resolve.

Finally, the wavefunctions for the first four states are plotted together with their corresponding energies and the confining potential (Fig. 5). The alternating parity, increasing node count, and quadratic growth of the energy levels all match the analytic infinite-well picture.

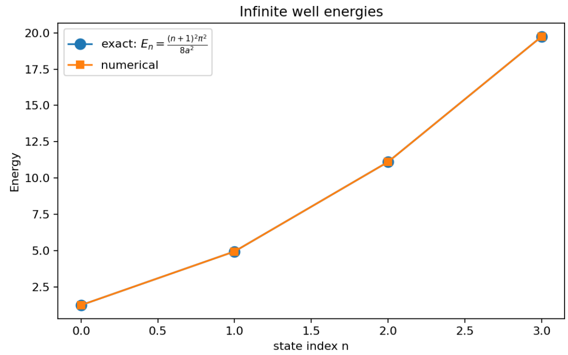


Figure 1: Comparison between numerical and exact energy eigenvalues for the infinite square well. The agreement is excellent for the first four states.

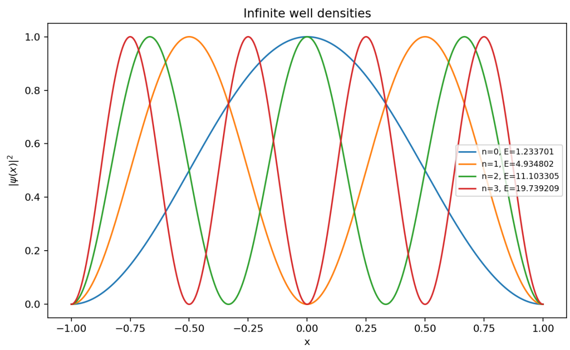
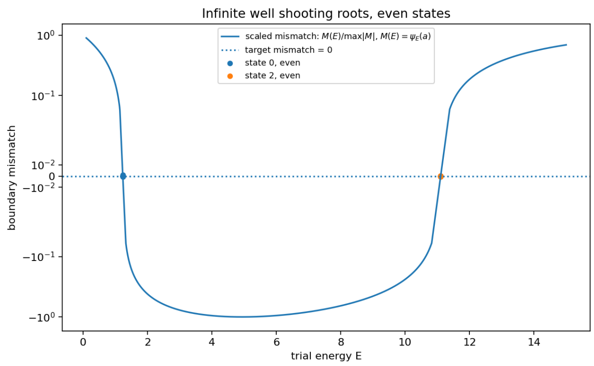
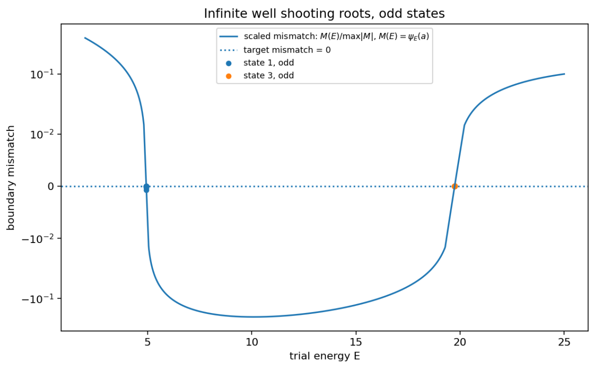


Figure 2: Probability densities $|\psi(x)|^2$ for the first four eigenstates of the infinite square well. The correct number of nodes and boundary behavior are clearly reproduced.



(a)



(b)

Figure 3: Boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues found by the shooting method.

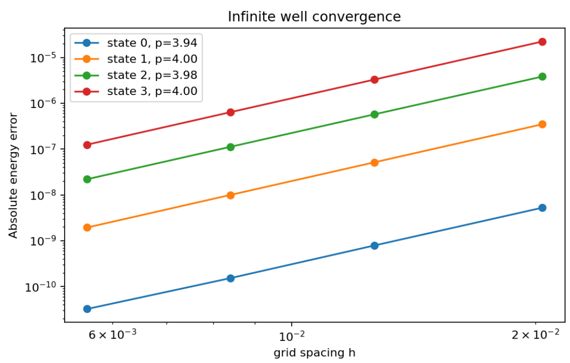


Figure 4: Convergence of the energy error as a function of grid spacing h. The slopes are close to fourth order, as expected for the Numerov method.

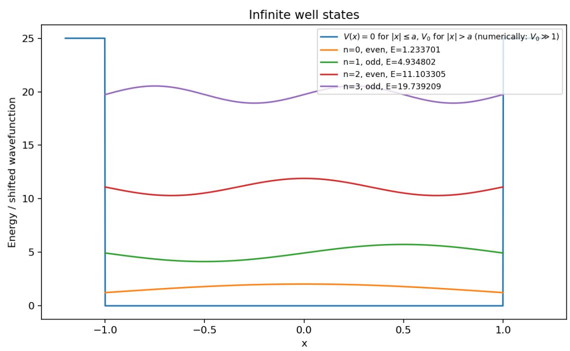


Figure 5: Wavefunctions for the first four states plotted together with their corresponding energies and the confining potential.

3.2 Harmonic oscillator

3.2.1 Numerov method on unbounded domains

For the harmonic oscillator with $V(x) = \frac{1}{2}\omega^2 x^2$, the exact spectrum is:

$$E_n = \omega\left(n + \tfrac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{5}$$

This provides a second analytic benchmark, but on an unbounded domain. Numerically, the domain is truncated to $[-x_{\max}, x_{\max}]$, and decaying boundary conditions are imposed.

The numerical eigenvalues agree with the exact linear spectrum to extremely high precision, as shown in Fig. 6. In the saved run the first four relative errors are 6.2×10^{-13}, 2.3×10^{-12}, 6.6×10^{-12}, and 7.5×10^{-12}. This is exactly what should happen for the harmonic oscillator benchmark: the inward-shooting formulation suppresses contamination by the exponentially growing forbidden-region solution and cleanly recovers the exact ladder $E_n = \omega(n + \frac{1}{2})$.

The probability densities (Fig. 7) show the expected structure: the ground state is centered and Gaussian-like, while the excited states alternate in parity and gain one additional node at each step. The support also broadens with increasing energy, consistent with the classical turning points moving outward. This confirms both physical correctness and stability of the integration on the truncated domain.

The shooting mismatch functions (Fig. 8) exhibit clear zero crossings at the expected energies. In the even sector the roots lie near $E \approx 0.5$ and $E \approx 2.5$, while in the odd sector they lie near $E \approx 1.5$ and $E \approx 3.5$. The mismatch spans many orders of magnitude because the integration starts in the decaying tail, but the sign changes remain sharp and the roots alternate even–odd as they should.

Convergence with respect to grid spacing h (Fig. 9) is overall consistent with fourth-order behavior. The saved study gives fitted slopes $p \approx 5.30$, 3.97, 4.57, and 4.00 for the first four states. The two slopes above four are not evidence of true superconvergence: they occur only after the errors have already dropped to the 10^{-11}–10^{-12} range, where root-finding tolerance and floating-point effects distort a short log–log fit. The raw errors themselves still decrease in the expected high-order way, from about 3.2×10^{-9}–2.6×10^{-8} on the coarsest grid to 5.1×10^{-13}–2.6×10^{-11} on the finest grid.

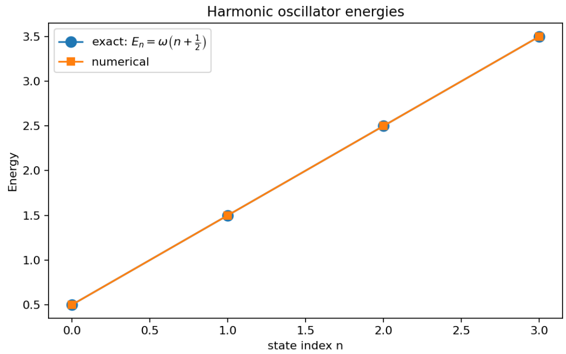
The dependence on domain size (Fig. 10) shows the expected truncation-to-discretization crossover. At $x_{\max} = 4$ the tail is clipped too aggressively and the fourth-state error is still about 5.9×10^{-5}. By $x_{\max} = 5$ the errors have already fallen dramatically, and by $x_{\max} \approx 6$ the curves flatten near the discretization floor: roughly 10^{-12} for the ground state, a few 10^{-12} for the first odd state, and about 10^{-11}–10^{-10} for the higher states. This confirms that the dominant error for large boxes is no longer domain truncation.

Finally, the combined state plot (Fig. 11) shows the wavefunctions together with the confining potential. The increasing spatial extent and alternating parity agree with the analytic harmonic-oscillator picture.

3.2.2 Numerov versus Runge–Kutta

As an additional method comparison, the harmonic oscillator was also solved using a fourth-order Runge–Kutta (RK4) shooting method. In this formulation the second-order Schrödinger equation is rewritten as the first-order system:

$$\psi' = \phi, \qquad \phi' = 2[V(x) - E]\psi \tag{6}$$



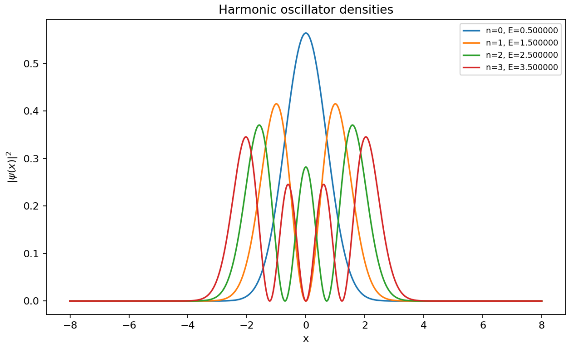
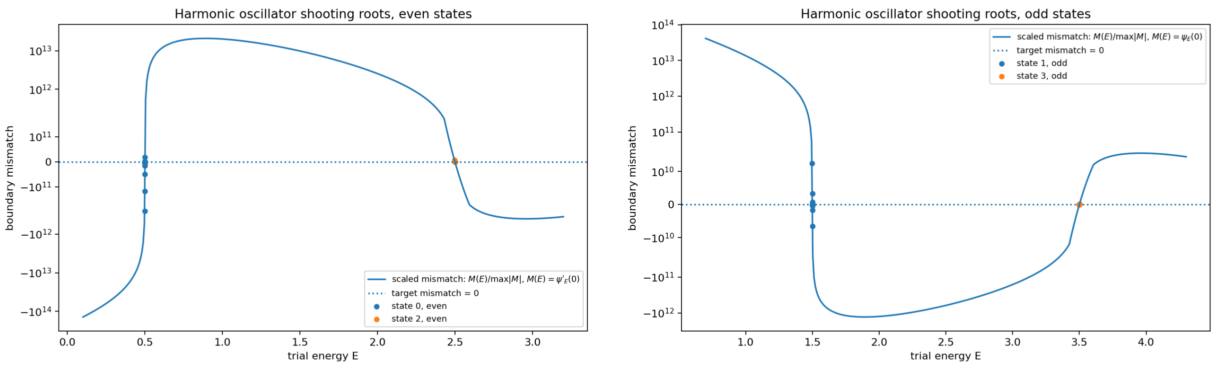


Figure 6: Comparison between numerical and exact energy eigenvalues for the harmonic oscillator. The agreement is excellent for the first four states.

Figure 7: Probability densities $|\psi(x)|^2$ for the first four harmonic oscillator eigenstates. The correct node structure is clearly reproduced.



(a) **(b)**

Figure 8: Boundary mismatch functions for even (a) and odd (b) states. The zero crossings correspond to the physical eigenvalues.

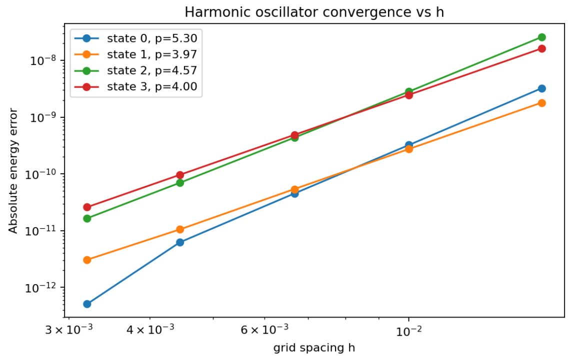


Figure 9: Convergence of the energy error as a function of grid spacing h. The slopes are close to fourth order, as expected for the Numerov method.

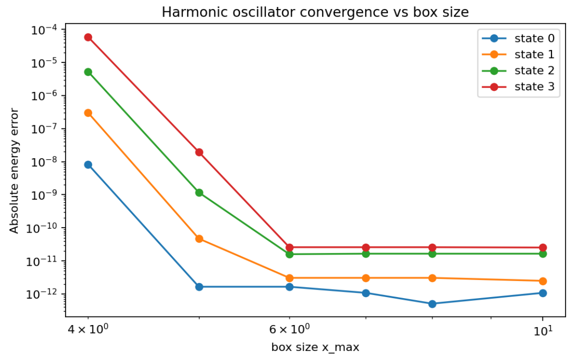


Figure 10: Dependence of the energy error on the domain size x_{max}. Errors decrease rapidly before reaching a plateau set by discretization and floating-point limits.

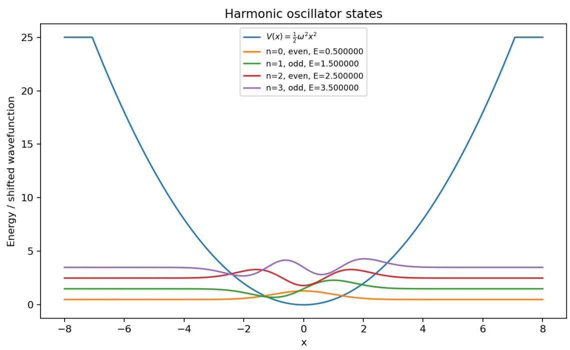


Figure 11: Wavefunctions for the first four harmonic oscillator states plotted together with the potential.

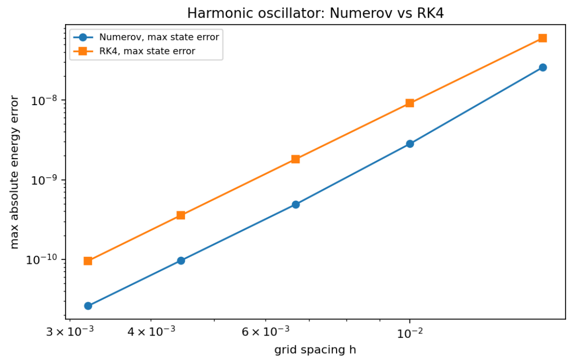


Figure 12: Comparison of Numerov and RK4 shooting for the harmonic oscillator. Both methods show fourth-order convergence, while Numerov gives a smaller maximum energy error at the same grid spacing.

This makes RK4 a useful general-purpose comparison for Numerov, which instead exploits the special second-order structure of the Schrödinger equation directly.

Figure 12 compares the maximum energy error over the computed harmonic-oscillator states as a function of grid spacing. Both methods show approximately fourth-order convergence, as expected. The RK4 slopes are much more uniformly close to four than the Numerov fit, with saved values $p \approx 4.04$, 4.02, 4.01, and 4.00. On the saved grids, the maximum absolute error ranges from 2.6×10^{-8} down to 2.6×10^{-11} for Numerov and from 6.0×10^{-8} down to 9.6×10^{-11} for RK4, so Numerov is consistently better by a factor of about 2–4 at fixed h even though both methods are fourth order.

The box-size study tells the same story for both methods. At fixed spacing, increasing $x_{\max}$ from 4 to 6 sharply reduces the truncation error, after which both methods level off at their respective discretization floors. This is a useful reminder that method order alone is not the whole story: domain truncation, root-finding tolerance, and boundary diagnostics all matter. After using a high-order boundary derivative for the Numerov inward-shooting condition, the Numerov and RK4 comparison behaves as expected and gives Numerov the smaller error at matched grid spacing.

3.3 Double-well potential

The quartic double-well potential is used as the main nontrivial bound-state example. In the shifted form used here:

$$V(x) = ax^4 - bx^2 - V_{\min}, \qquad V_{\min} = -\frac{b^2}{4a} \tag{7}$$

the two minima are shifted to zero. Unlike the infinite square well and harmonic oscillator, this system does not have a simple analytic spectrum, so the validation relies on internal consistency checks, convergence studies, parity structure, and physically expected tunneling behavior.

Figure 13 shows the first four computed states together with the double-well potential. The two lowest states form a nearly degenerate even-odd pair, with energies $E_0 \approx 2.357273$ and $E_1 \approx 2.359372$. This is the expected tunneling doublet: the even and odd states are symmetric and antisymmetric combinations of states localized in the two wells. The next pair is higher in energy and has a larger splitting, consistent with stronger overlap through the barrier.

The probability densities in Fig. 14 show the expected localization near the two wells. The lowest pair has nearly identical density, because the sign difference between even and odd wavefunctions disappears in $|\psi(x)|^2$. The higher states show additional node structure and broader support, as expected for excited double-well states.

The root-finding diagnostics (Fig. 15) show well-defined zero crossings in the even and odd parity sectors. These plots verify that the computed energies are roots of the shooting mismatch function and that the parity separation continues to work for a more complicated potential.

Since there is no closed-form exact spectrum for this potential, grid convergence is measured using successive refinements at fixed domain size. The convergence with respect to grid spacing (Fig. 16) still shows stable high-order behavior, but the fitted slopes span $p \approx 4.18$, 3.95, 4.55, and 5.67 for the first four states. As in the harmonic oscillator case, the values above four occur when the successive-difference errors are already very small, so the figure should be interpreted as evidence of stable refinement rather than literal fifth-order convergence.

The box-size study (Fig. 17) shows rapid error reduction as the computational domain is enlarged, followed by a small residual floor relative to the larger-box reference solution. This confirms that the chosen domain is large enough for the bound states and that the final results are not dominated by artificial boundary truncation.

Finally, Fig. 18 shows how the lowest even-odd splitting changes with the double-well parameter b. As b increases, the barrier between the wells becomes more pronounced and the splitting decreases toward zero. This is the expected physical signature of tunneling suppression: increasing the barrier reduces overlap between the left- and right-localized components.

Overall, the double-well results demonstrate that the solver works beyond analytic benchmarks. The computed states have the expected parity and localization, the tunneling splitting behaves physically, and the convergence studies show that the numerical results are stable with respect to both grid spacing and domain size.

3.4 Finite square well

The finite square well is included as an additional nontrivial bound-state test. Unlike the infinite square well, the potential walls are finite, so bound states can leak into the classically forbidden region. The potential used here has the form:

$$V(x) = \begin{cases} 0 & , |x| \leq a \\ V_0 & , |x| > a \end{cases} \tag{8}$$

and only states with $E < V_0$ are true bound states.

Figure 19 shows the first four computed states together with the finite well potential. The first three states lie clearly below the barrier and are localized mainly inside the well, with exponentially decaying tails outside. The fourth state is close to the barrier height, and therefore has much stronger leakage into the exterior region. This behavior is physically expected for shallow or near-threshold bound states.

The probability densities in Fig. 20 show the same effect more clearly. Low-lying states are concentrated inside the well, while the highest plotted state is spread much farther outside the well. The node count and even-odd structure remain consistent with the ordering of one-dimensional bound states.

These results are sufficient for the finite square well part of the report because this case is used mainly to demonstrate that the same shooting framework works for a potential with finite barriers and evanescent tails. The more detailed convergence and validation analysis is already provided by the infinite well and harmonic oscillator, while the finite well serves as an additional qualitative and physical test.

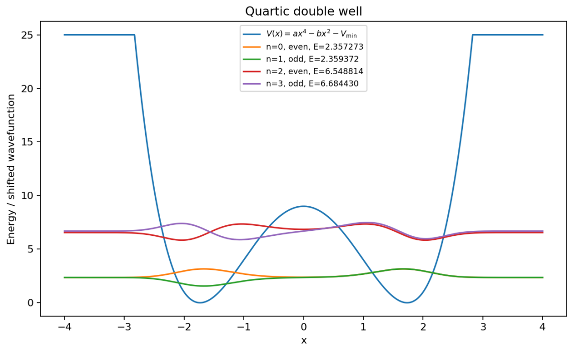


Figure 13: First four bound states of the quartic double-well potential. The lowest two states form a nearly degenerate tunneling doublet.

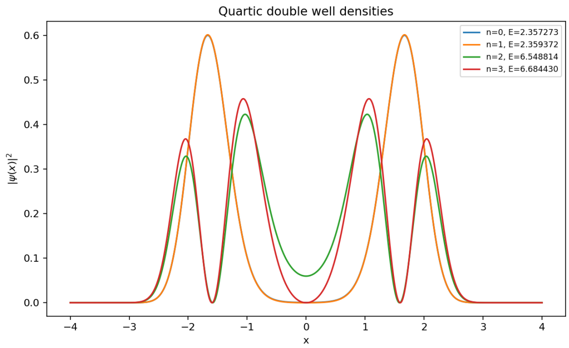
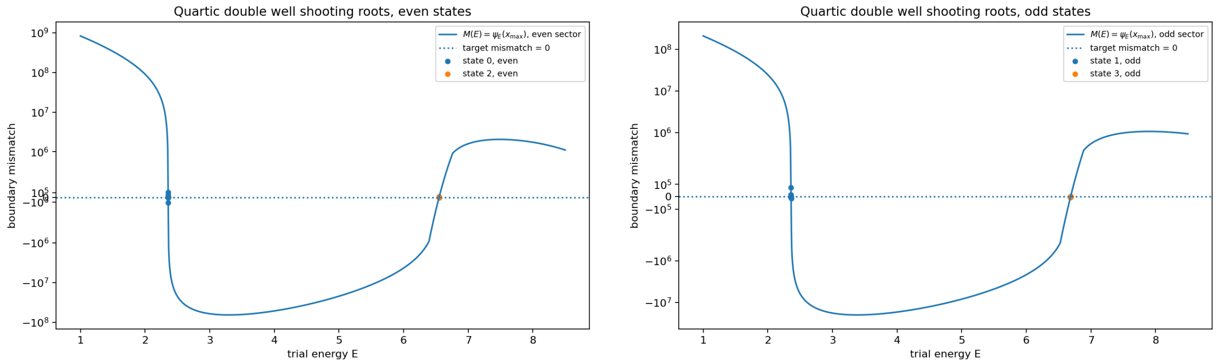


Figure 14: Probability densities $|\psi(x)|^2$ for the first four double-well states. The lowest even and odd states have nearly identical densities, as expected for a tunneling doublet.



(a) **(b)**

Figure 15: Shooting mismatch functions for even (top) and odd (bottom) double-well states. The zero crossings define the numerical eigenvalues.

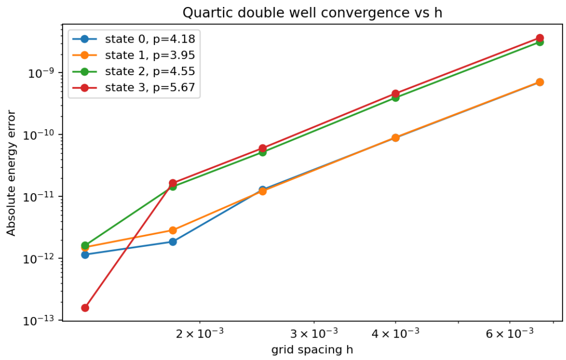


Figure 16: Grid convergence for the quartic double well. The energy errors decrease with high-order behavior under successive grid refinement.

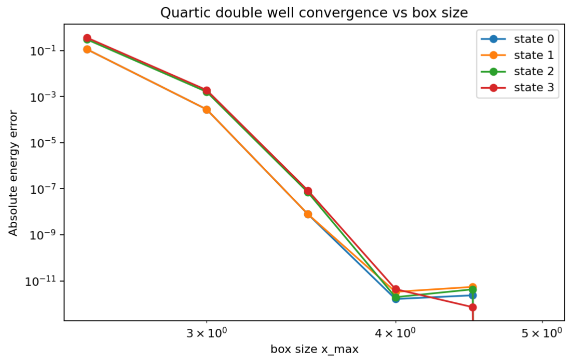


Figure 17: Dependence of the double-well energy errors on box size. The errors decrease rapidly as the domain is enlarged and then reach a numerical plateau.

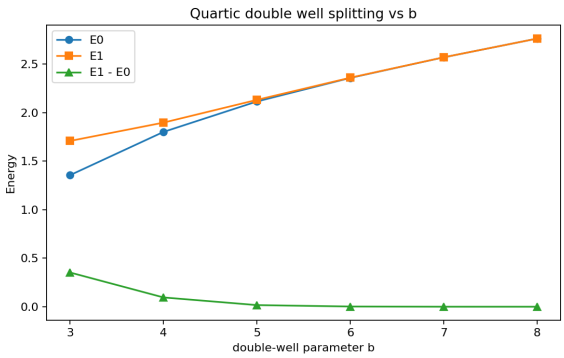


Figure 18: Energy splitting between the two lowest double-well states as the parameter b is varied. The splitting decreases as the barrier becomes stronger, consistent with tunneling suppression.

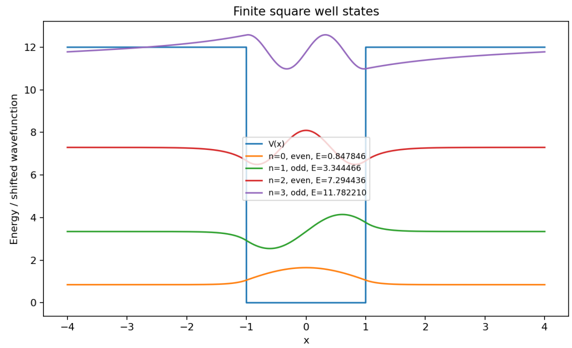


Figure 19: First four finite square well states plotted together with the finite barrier potential. States below the barrier are localized in the well, while the near-threshold state has a larger evanescent tail.

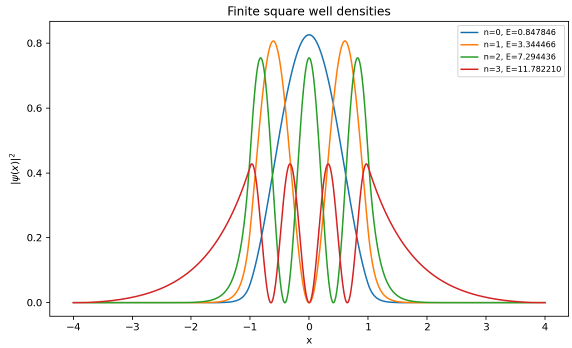


Figure 20: Probability densities $|\psi(x)|^2$ for the first four finite square well states. The densities show both interior oscillation and exterior exponential leakage.

3.5 Scattering and resonant tunneling

3.5.1 Single barrier potential

As a secondary extension beyond bound states, the Numerov framework is also applied to one-dimensional scattering from a finite rectangular barrier. For each incident energy E, the complex wavefunction is propagated through the barrier region and decomposed into incident, reflected, and transmitted waves. This gives the transmission and reflection probabilities $T(E)$ and $R(E)$.

The single-barrier result is shown in Fig. 21. At low incident energy, where E is well below the barrier height, transmission is strongly suppressed and reflection is close to one. As the incident energy increases, the transmission rises smoothly and approaches unity, while reflection decreases. This is the expected qualitative behavior for scattering from a finite barrier.

The dashed curve shows the conservation check $T(E) + R(E)$, which is numerically indistinguishable from one across the full energy range in the saved data: the largest deviation is about 2.3×10^{-12}. This confirms that the numerical scattering calculation conserves probability flux to within floating-point error for this setup.

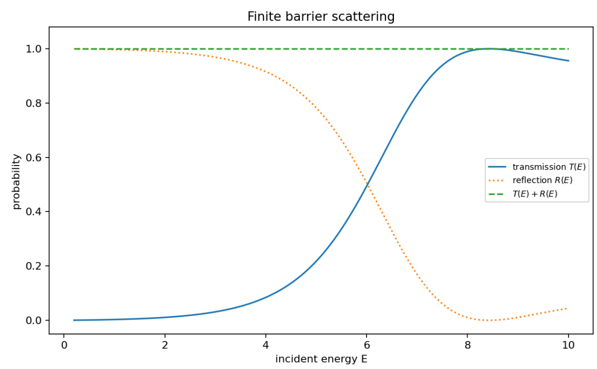


Figure 21: Transmission and reflection probabilities for scattering from a finite rectangular barrier. At low energies reflection dominates, while at high energies transmission approaches unity. The conservation check $T(E) + R(E)$ remains close to one.

3.5.2 Double barrier potential

The double-barrier potential is used to demonstrate resonant tunneling. In this case the two barriers create an intermediate well, so certain incident energies can form quasi-bound states between the barriers. At these energies, destructive reflection and constructive transmission lead to sharp peaks in the transmission coefficient.

Figure 22 shows the computed transmission and reflection probabilities. The transmission contains a narrow resonance near $E \approx 1.139$ and a broader high-energy resonance at larger energy. At the resonant energies, $T(E)$ approaches unity and $R(E)$ drops correspondingly, which is the expected behavior for resonant tunneling through a double barrier.

The conservation check $T(E) + R(E)$ is again essentially exact in the saved calculation, with a largest deviation of about 8.4×10^{-12}. This is an important diagnostic because it verifies that the numerical decomposition into incident, reflected, and transmitted components is self-consistent.

Figure 23 shows the probability density at the first resonance. The wavefunction is strongly enhanced in the region between the two barriers, which is the expected signature of a quasi-bound state. This spatial localization explains the corresponding peak in the transmission spectrum.

These results are sufficient for the double-barrier part of the report because the scattering section is a secondary extension. The plots demonstrate the key qualitative physics: transmission resonances, reflection suppression at resonance, flux conservation, and enhancement of the wavefunction in the central well.

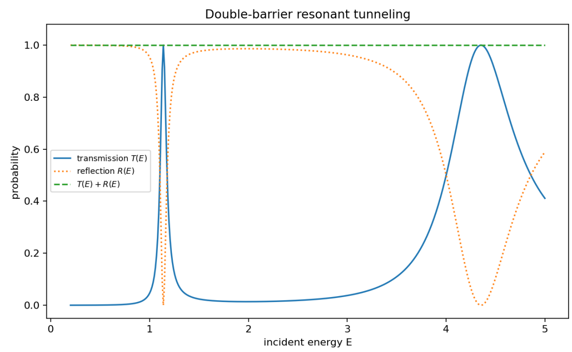


Figure 22: Transmission and reflection probabilities for the double-barrier potential. Sharp transmission peaks occur at resonant energies, where reflection is suppressed and $T(E) + R(E)$ remains close to one.

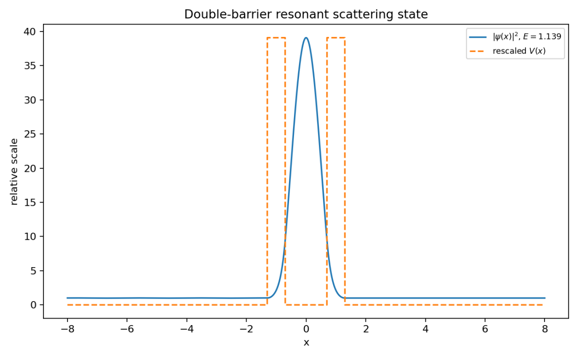


Figure 23: Probability density at the first resonant energy. The wavefunction is strongly enhanced between the two barriers, showing the quasi-bound state responsible for resonant tunneling.

4 Discussion

The calculation was checked in four complementary ways. First, an automated validation suite tests normalization, the fourth-order boundary derivative, analytic benchmark energies, the double-well level ordering, scattering flux conservation, and the entry-script import path. Second, the infinite square well and harmonic oscillator reproduce known analytic spectra, providing direct validation of the eigenvalue search. Third, the wavefunctions have the expected parity, node structure, normalization, and boundary behavior. Fourth, convergence tests with respect to grid spacing and domain size show that the results are stable under numerical refinement.

The convergence behavior also exposes important numerical details. In the infinite square well, the observed fourth-order convergence confirms that the Numerov recurrence, startup conditions, and boundary treatment are consistent. For the harmonic oscillator, the box-size study separates domain truncation error from discretization error: increasing $x_{\max}$ initially reduces the error rapidly, after which the error reaches a plateau set by grid spacing, root-finding tolerance, and floating-point limits. Slopes slightly above fourth order in some fits are therefore interpreted as small-error fitting artifacts rather than evidence of a truly higher-order method.

The comparison with RK4 shows why method order alone is not enough. RK4 evolves both ψ and ψ' directly, while Numerov requires numerical boundary diagnostics. Once the boundary derivative in the Numerov inward-shooting calculation is evaluated with comparable accuracy, Numerov behaves as expected and gives smaller errors than RK4 at the same grid spacing for the harmonic oscillator.

For the quartic double well, where no closed-form spectrum is available, correctness is established through self-consistency rather than analytic comparison. The use of an analytic potential shift, separated grid and box convergence studies, parity-specific root diagnostics, and normalized boundary leakage all support the reliability of the computed states. The decreasing even-odd splitting with increasing barrier parameter is the expected physical signature of tunneling suppression.

The scattering examples are treated as a secondary extension. The single-barrier calculation shows the expected transition from reflection-dominated behavior at low energy to transmission-dominated behavior at high energy, while the double-barrier calculation shows resonant transmission peaks and enhanced probability density in the central well. In both cases, the saved calculations satisfy $T(E) + R(E) = 1$ to within about 10^{-11}–10^{-12}, providing an additional numerical consistency check.

5 Conclusion

This project implemented a Numerov shooting solver for the one-dimensional time-independent Schrödinger equation. The method was validated on analytic bound-state problems, extended to finite and double-well potentials, and checked through convergence studies, parity diagnostics, and comparison with RK4. The results reproduce known spectra, show the expected tunneling splitting in the double well, and demonstrate resonant transmission in scattering as a secondary extension.

References

[1] Tao Pang. *An Introduction to Computational Physics*. 2nd ed. Cambridge University Press, 2006.